

# Probability Basics

## Video Lesson

Duration: 00:12:46

## Slide Deck

**M11S03** (<https://canvas.tamu.edu/courses/430127/files/92570689?wrap=1>) 

## Lesson Transcript

In this lecture, we will look into probability basics.

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## Probability Notations

- **Random variable**: variable that can take on different values
  - $A, B, \dots$ : boolean values (**True** or **False** ).
  - $X, Y, \dots$  : numerical values or other multi-valued enumerations (1, 2, 0.5, Cloudy, Rainy, Sunny, ...)
- $P(X = v)$  : **probability** of the variable  $X$  having value  $v$ .
  - This can be viewed as an *event*.
  - For boolean variables,  $P(A)$  means  $P(A = \text{True})$ , and  $P(\neg A)$  means  $P(A = \text{False})$ .
- $\mathbf{P}(X)$  : **probability distribution**, a full list of probabilities for all possible values that  $X$  can take (note that **P** is in **bold**).

First, let us look at some probability notations. These are random variables, probability, and probability distributions.

A random variable is a variable that can take different values. For example, they can take boolean values **True** or **False**, and let us give them a name,  $A, B, C, D$ , and so on. On the other hand, they can also take on different numerical values or other multi-valued enumerations. For example, 1, 2, 0.5, Cloudy, Rainy, Sunny, etc.

And let us say we can give them a name,  $X, Y, Z$ , etc.

The probability  $P(X = v)$  is the probability of that variable  $X$  having a value of  $v$ . This can be viewed as an event. For boolean variables,  $P(A)$  can be used to represent  $P(A = \text{True})$ , and  $P(\neg A)$  to represent  $P(A = \text{False})$ .

$\mathbf{P}(X)$ , where we write this **P** in **bold**, represents the probability distribution. This is a full list of probabilities for all possible values that  $X$  can take.

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## Examples

- Boolean:  
 $P(\text{Infected}) = 0.01, P(\neg \text{Infected}) = 0.99.$
- Multi valued:  $\mathbb{I}$   
 $P(\text{Dice} = 1) = \frac{1}{6}, P(\text{Dice} = 2) = \frac{1}{6}, \dots$
- Multi valued:  
 $P(\text{Weather} = \text{Sunny}) = 0.7,$   
 $P(\text{Weather} = \text{Rainy}) = 0.2, \dots$

For example, for boolean random variables *Infected*,  $P(\text{Infected})$  would be 0.01.  $P(\neg \text{Infected})$  would be 0.99. So, this is the same as  $P(\text{Infected})$  equals **True**, and this  $P(\text{Infected})$  equals **False**.

For multi-valued numerical values, we have an example the probability of *Dice* equals 1 is 1/6. The probability of *dice* equals two is 1/6, and so on. Or we can have a multi-valued variable or random variable with symbolic values such as *Sunny*, *Rainy*, or *Cloudy*. In this case, the random variable would be *weather* with the probability that the weather is sunny and it is 0.7, let us say, in Texas. Maybe this is not the case in Washington State.

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## Logical Connectives and Conditional Probability

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- Logical connectives can be used:

$$P(A \vee B), P(A \wedge \neg B), P(\text{Cavity} \wedge \neg \text{Insured}), \text{ etc.}$$

- Conditional Probability  $P(A|B)$  (read *probability of A given B*):

$$P(\text{Cavity} | \text{Toothache}) = 0.8$$

- As new evidence comes in, the conditional probability gets updated:

$$P(\text{Cavity} | \underbrace{\text{Toothache} \wedge \text{BadBreath}})$$

With boolean random variables, we can use logical connectives, and also we can consider conditional probability. Of course, conditional probability can be used for other random variables, not just boolean.

Let us look at this example where logical connectives OR, or disjunction, AND or conjunction, NOT or negation, and these are used in combination. So, what is the probability of  $A \vee B$ ? What is

the probability of  $A$  and  $\neg B$ ? What is the probability of a cavity and not being insured? And so on.

Conditional probability is written as  $P(A | B)$ , and actually, in full, it is read probability of  $A$  given  $B$ . So, here is an example. What is the probability of a cavity given toothache? Let us say it has some kind of numerical value. Let us say this is the only information that you have, toothache, but let us say you also have bad breath.

Then that becomes new evidence, and you can combine that with the existing evidence by using this conjunction.

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## Logical Connectives and Conditional Probability

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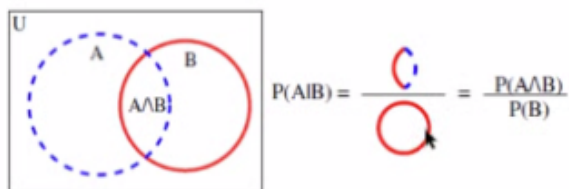
- As new evidence comes in, the conditional probability gets updated:

$$P(\text{Cavity} | \underbrace{\text{Toothache} \wedge \text{BadBreath}}) = 0.9$$

Once you have this new information, perhaps your belief about the probability of a cavity would actually go up compared to the previous situation where you had just one piece of evidence, which is a toothache.

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# Conditional Probability



- Think about the **area** occupied by each event.
- The bounding rectangle  $U$  has an area of 1, thus

$$P(A) = \frac{\text{Area of } A}{\text{Area of } U} = \frac{\text{Area of } A}{1} = \text{Area of } A$$

- $P(A|B)$  means  $B$  now takes on the role of  $U$ . Within this limited event space, what is the probability of  $A$ .

Conditional probability. The probability of  $A$  given  $B$  is defined as probability  $A$  and  $B$  divided by probability  $B$ .

That is equivalent to saying, what is the probability of  $A$  when we only consider events that satisfy this boolean random variable  $B$ ? To compute that, we can just divide this area where  $A$  and  $B$  co-occur by the area where these  $B$  events occur.

One other way of expressing that is  $P(A | B)$  means  $B$  now taking on the role of the universe here, and within this limited event space, what is the probability of  $A$ ?

## The Axioms of Probability

All axioms

- 1 All probabilities are between 0 and 1

$$0 \leq P(A) \leq 1$$

- 2 For a valid proposition  $A$  (**True** under all interpretations):  $P(A) = 1$ , and for a inconsistent proposition  $A$  (**False** under all interpretations):  $P(A) = 0$ .

- 3  $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

Other properties follow from these three axioms.

There are three axioms of probability.

First, all probabilities are between 0 and 1. So, the probability value should be 0 or 1 or anywhere

in between.

For a valid proposition,  $A$ , that is **True** on all interpretations,  $P(A = 1)$ . Again, here we are considering  $A$  to be a boolean random variable, and for an inconsistent proposition  $A$ , that is, it is **False** under all interpretations. Then that probability should be 0.

Then, we have this disjunction rule: probability  $A \vee B$  equals the probability of  $A$  plus the probability of  $B$  minus the probability of the conjunction of  $A$  and  $B$ .

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## Other Properties

- From the axioms,

$$\begin{aligned}P(A \vee \neg A) &= P(A) + P(\neg A) - P(A \wedge \neg A) \\P(\mathbf{True}) &= P(A) + P(\neg A) - P(\mathbf{False}) \\1 &= P(A) + P(\neg A) \\P(\neg A) &= 1 - P(A)\end{aligned}$$

- More generally, the **sum** of probabilities  $P(X = v)$  is 1, for all values  $v$  the random variable  $X$  can take:

$$\left[ \sum_{v \in V} P(X = v) \right] = 1,$$

where  $V$  is the set of all possible values  $X$  can take.

From that last axiom, let us plug-in  $A$  and  $\neg A$  in place of  $A$  and  $B$ . Then the left-hand side becomes probability  $A \vee \neg A$ . The right-hand side becomes the probability of  $A$  plus probability of  $\neg A$  minus probability  $A \wedge \neg A$ . This evaluates to true, and that evaluates to **False**. So, let us plug in these values.

From the second axiom, we know the  $P$  of **True** is 1, and the  $P$  of **False** is 0. Hence, this disappears, and we get 1 equals  $P(A)$  plus  $P(\neg A)$ .

Finally, subtracting this and that with  $P(A)$ , we get 1 minus  $P(A)$  and  $P(\neg A)$ .

More generally, for multi-value random variables, the sum of the probability of  $X$  equaling a specific value over all possible values should add up to 1.

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# Joint Probability Distribution

For random variables  $X_1, X_2, \dots, X_n$ ,

- An **atomic event** is an assignment of particular values to each random variable.
- The **joint probability distribution**  $P(X_1, X_2, \dots, X_n)$  completely specifies the probabilities of all **atomic events**.
- Thus,

$$\left[ \sum_{(v_1, v_2, \dots, v_n) \in \mathbf{V}} P(X_1 = v_1, X_2 = v_2, \dots, X_n = v_n) \right] = 1,$$

where  $\mathbf{V}$  is a set of all possible  $n$ -vectors that the vector  $(X_1, X_2, \dots, X_n)$  can assume..

Next, let us look at joint probability distribution. Let us say we have  $n$  different random variables. An atomic event assigns particular values to each of these random variables. If we have a boolean random variable, then we have  $2^n$  different atomic events that are possible. The joint probability distribution  $P$  of  $X_1, X_2, \dots, X_n$  completely specifies the probabilities of all atomic events.

## Joint Probability Distribution

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where  $\mathbf{V}$  is a set of all possible  $n$ -vectors that the vector  $(X_1, X_2, \dots, X_n)$  can assume..

| $X_1$ | $X_2$ | $P(X_1, X_2)$ |
|-------|-------|---------------|
| T     | T     | 0.6           |
| T     | F     | 0.1           |
| F     | T     | 0.12          |
| F     | F     | 0.18          |

For example, let us consider this joint probability distribution with two random variables,  $X_1$  and  $X_2$ . Let us say these two are boolean random variables. So, in this case, we have 1, 2, 3, and 4 atomic events where probability  $X_1$  equals **True**, and  $X_2$  equals **True** is 0.6.  $X_1$  equals **True**,  $X_2$  equals **False** is 0.1, and so on. And then we add all of these up, and then we get 1.0.

As we can see, when we have two random variables, we need  $2^2$  atomic events to fully describe the joint probability distribution. This equation here shows the same thing for the general case where we have  $n$  different random variables in the joint probability distribution.

## Joint Probability Distribution: Example

|               | Toothache     | $\neg$ Toothache   | Sum               |
|---------------|---------------|--------------------|-------------------|
| Cavity        | 0.04          | 0.06               | $P(C) = 0.1$      |
| $\neg$ Cavity | 0.01          | 0.89               | $P(\neg C) = 0.9$ |
| Sum           | $P(T) = 0.05$ | $P(\neg T) = 0.95$ | $\sum = 1.0$      |

Abbreviations:  $C$  = Cavity,  $T$  = Toothache

- $P(C \vee T) = P(C) + P(T) - P(C \wedge T)$   
 $= 0.1 + 0.05 - 0.04 = 0.11$
- $P(C|T) = \frac{P(C \wedge T)}{P(T)} = \frac{0.04}{0.05} = 0.8$
- $P(T|C) = \frac{P(C \wedge T)}{P(C)} = \frac{0.04}{0.1} = 0.5$

| Tooth | Cav | $P(\text{Tooth}, \text{Cav})$ |
|-------|-----|-------------------------------|
| T     | T   | 0.04                          |
| T     | F   | 0.01                          |
| F     | T   | 0.06                          |
| F     | F   | 0.89                          |

In practice, writing a full joint probability table like this is impossible (or too much effort): for  $n$  boolean random variables, you need  $2^n$  entries.

Let us look at this example. You can either have a toothache or not a toothache. You can either have a cavity or no cavity. When you have a toothache and a cavity, let us say the probability is 0.04. You can write this table in this format where each row now is an atomic event.

When the toothache is **True**, and a cavity is **True**, that corresponds to this, so the probability or the joint probability would now become 0.04 for that.

In the case of a toothache and not a cavity, you can also look it up in this table on the right with the probability of a toothache and not having a cavity. Then it is 0.01, which corresponds to that value, etc. For the rest, you can determine which item in this left table matches which row on the right-hand side.

From this, we can compute things like this. What is the probability of a cavity or a toothache? We can use the axiom number 3 to compute this. What is the probability of a cavity plus the probability of a toothache minus the probability of a cavity and a toothache?

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| Tooth | Cav | $P(\text{Tooth}, \text{Cav})$ |
|-------|-----|-------------------------------|
| T     | T   | 0.04                          |
| T     | F   | 0.01                          |
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| F     | F   | 0.89                          |

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Here we can simply look up this value from the table here or here when there is a cavity and when there is a toothache. Okay, but what about these two values? There is a probability of a cavity. There is a probability of a toothache.

## Joint Probability Distribution: Example

|               | Toothache     | $\neg$ Toothache   | Sum               |
|---------------|---------------|--------------------|-------------------|
| Cavity        | 0.04          | 0.06               | $P(C) = 0.1$      |
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| Tooth | Cav | $P(\text{Tooth}, \text{Cav})$ |
|-------|-----|-------------------------------|
| T     | T   | 0.04                          |
| T     | F   | 0.01                          |
| F     | T   | 0.06                          |
| F     | F   | 0.89                          |

In practice, writing a full joint probability table like this is impossible (or too much effort): for  $n$  boolean random variables, you need  $2^n$  entries.

For the probability of a toothache, you can simply add these two values to get 0.05. Also, you can look at it in the joint probability table on the right-hand side. So, where there is a true value on the toothache, you just add the rows, and you get the probability. Likewise, for the cavity, you just add these values.

Thus, you pick those rows where a cavity is **True** and add the values. Then you get the desired result, which is 0.1. For that, the result is 0.05. So, when you plug in these numbers, you get 0.11 as the probability of a cavity or a toothache.

## Joint Probability Distribution: Example

|               | Toothache     | $\neg$ Toothache   | Sum               |
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In practice, writing a full joint probability table like this is impossible (or too much effort): for  $n$  boolean random variables, you need  $2^n$  entries.

We can compute all of these condition probabilities. What is the probability of a cavity given toothache? Then using the definition, this would be the probability of cavity and toothache divided by the probability of toothache. So, looking this up from the table 0.04, and then looking this up from this computed value, this becomes 0.8.

The opposite. What is the probability of toothache given a cavity? Again, we look up the joint probability and divide it by this probability here, where this is called the marginal probability. Then we get 0.04 divided by 0.1, and we get a value of 0.5.

Of course, we can use this kind of method to compute any arbitrary, let us say, probabilities like that or conditional probabilities. But in practice, writing up all of these rows in the full joint probability table is very inefficient because for joint probability table with  $n$  random variables in it needs  $2^n$  entries.

In the following lecture, we will look at the Bayes Rule.